

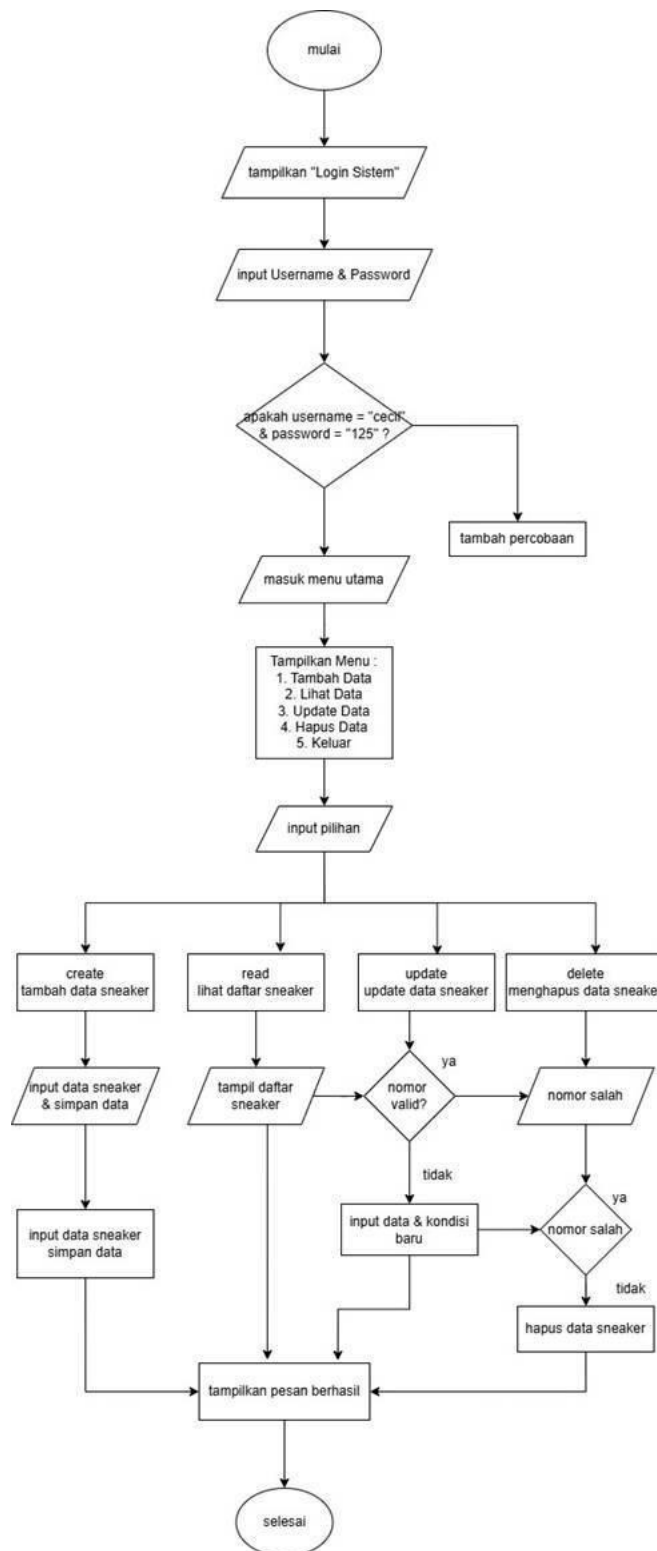
LAPORAN PRAKTIKUM
POSTTEST 5
ALGORITMA PEMROGRAMAN LANJUT



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PROGRAM STUDI INFORMATIKA
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1. Flowchart



Gambar1.1 Flowchart

2. Deskripsi Singkat Program

Program ini digunakan untuk mengelola data koleksi sneaker menggunakan konsep CRUD (Create, Read, Update, Delete). Program ini dirancang untuk menyimpan dan mengelola informasi sneaker seperti brand, model, warna, ukuran, harga, dan kondisi sepatu. Untuk mengelola data sneaker yang dilengkapi dengan fitur **login** dan **sorting (pengurutan data)**.

Login User

- Pengguna harus memasukkan username dan password
- Maksimal 3 kali percobaan

Menampilkan Data

- Menampilkan seluruh data sneaker yang tersedia

Sorting	Data	(Sesuai	Modul)
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Program menerapkan 3 metode sorting, yaitu:

- **Bubble Sort (Ascending)**

Mengurutkan data berdasarkan brand (A–Z)

- **Selection Sort (Descending)**

Mengurutkan data berdasarkan harga (tertinggi ke terendah)

- **Insertion Sort (Ascending)**

Mengurutkan data berdasarkan ukuran (kecil ke besar)

3. Source Code

```
#include <iostream>
using namespace std;

struct DetailSneaker{
    string warna;
    int ukuran;
};

struct Sneaker{
    string brand;
    string model;
    int harga;
    string kondisi;
    DetailSneaker detail;
};

struct User{
    string username;
    string password;
};

Sneaker dataSneaker[100];
int jumlahData = 0;
User akun = {"cecil","125"};

bool login(User user){
    string username, password;
    int percobaan = 0;

    while(percobaan < 3){
        cout << "\n=== LOGIN ===\n";
        cout << "Username : "; cin >> username;
        cout << "Password : "; cin >> password;
```

```

        if(username == user.username && password == user.password){
            cout << "Login berhasil!\n";
            return true;
        } else {
            cout << "Login gagal!\n";
            percobaan++;
        }
    }
    return false;
}

void isiDataAwal(Sneaker *data, int &jumlah){
    data[0] = {"Nike","AirForce1",1500000,"Baru",{"White",42}};
    data[1] = {"Adidas","Yeezy350",4200000,"Baru",{"Black",43}};
    data[2] = {"Converse","ChuckTaylor",850000,"Bekas",{"Red",41}};
    jumlah = 3;
}

void tampilData(Sneaker *data, int jumlah){
    cout << "\n=== Data Sneaker ===\n";
    for(int i=0; i<jumlah; i++){
        cout << i+1 << ". "
            << data[i].brand << " "
            << data[i].model << " | "
            << data[i].detail.warna << " "
            << data[i].detail.ukuran << " | "
            << data[i].harga << " | "
            << data[i].kondisi << endl;
    }
}

void bubbleSortBrand(Sneaker data[], int jumlah){
    for(int i=0; i<jumlah-1; i++){
        for(int j=0; j<jumlah-i-1; j++){
            if(data[j].brand > data[j+1].brand){
                swap(data[j], data[j+1]);
            }
        }
    }
}

```

```

}

void selectionSortHarga(Sneaker data[], int jumlah){
    for(int i=0; i<jumlah-1; i++){
        int indeksMaks = i;

        for(int j=i+1; j<jumlah; j++){
            if(data[j].harga > data[indeksMaks].harga){
                indeksMaks = j;
            }
        }

        swap(data[i], data[indeksMaks]);
    }
}

void insertionSortUkuran(Sneaker data[], int jumlah){
    for(int i=1; i<jumlah; i++){
        Sneaker kunci = data[i];
        int j = i-1;

        while(j >= 0 && data[j].detail.ukuran > kunci.detail.ukuran){
            data[j+1] = data[j];
            j--;
        }
        data[j+1] = kunci;
    }
}

void menu(){
    cout << "\n=== MENU ===\n";
    cout << "1. Tampil Data\n";
    cout << "2. Sorting\n";
    cout << "3. Keluar\n";
}

int main(){

    if(!login(akun)){
        cout << "Gagal login.\n";
        return 0;
    }
}

```

```

}

isiDataAwal(dataSneaker, jumlahData);

int pilihan;

do{
    menu();
    cout << "Pilih : ";
    cin >> pilihan;

    if(pilihan == 1){
        tampilData(dataSneaker, jumlahData);
    }

    else if(pilihan == 2){
        int pilihSorting;

        cout << "\n=== PILIH SORTING ===\n";
        cout << "1. Brand (Ascending - Bubble)\n";
        cout << "2. Harga (Descending - Selection)\n";
        cout << "3. Ukuran (Ascending - Insertion)\n";
        cout << "Pilih : ";
        cin >> pilihSorting;

        if(pilihSorting == 1){
            bubbleSortBrand(dataSneaker, jumlahData);
        }
        else if(pilihSorting == 2){
            selectionSortHarga(dataSneaker, jumlahData);
        }
        else if(pilihSorting == 3){
            insertionSortUkuran(dataSneaker, jumlahData);
        }

        tampilData(dataSneaker, jumlahData);
    }

}while(pilihan != 3);

return 0;

```

```
}
```

4. Hasil Output

```
===== LOGIN SISTEM =====  
Username : cecil  
Password : 125  
Login berhasil!
```

Gambar 4.1 Menu Login

```
=== MENU ===  
1. Tampil Data  
2. Sorting  
3. Keluar  
Pilih : 1
```

Gambar 4.2 Menu Utama

```
=== Data Sneaker ===  
1. Nike AirForce1 | White 42 | 1500000 | Baru  
2. Adidas Yeezy350 | Black 43 | 4200000 | Baru  
3. Converse ChuckTaylor | Red 41 | 850000 | Bekas
```

Gambar 4.3 Tampil Data (sebelum Sorting)

```
=== PILIH SORTING ===  
1. Brand (Ascending - Bubble)  
2. Harga (Descending - Selection)  
3. Ukuran (Ascending - Insertion)  
Pilih : 1  
  
=== Data Sneaker ===  
1. Adidas Yeezy350 | Black 43 | 4200000 | Baru  
2. Converse ChuckTaylor | Red 41 | 850000 | Bekas  
3. Nike AirForce1 | White 42 | 1500000 | Baru
```

Gambar 4.4 Sorting Huruf (Brand - Ascending / Bubble Sort)


```
=== PILIH SORTING ===
1. Brand (Ascending - Bubble)
2. Harga (Descending - Selection)
3. Ukuran (Ascending - Insertion)
Pilih : 2

=== Data Sneaker ===
1. Adidas Yeezy350 | Black 43 | 4200000 | Baru
2. Nike AirForce1 | White 42 | 1500000 | Baru
3. Converse ChuckTaylor | Red 41 | 850000 | Bekas
```

Gambar 4.5 Sorting Harga (Descending / Selection Sort)

```
=== PILIH SORTING ===
1. Brand (Ascending - Bubble)
2. Harga (Descending - Selection)
3. Ukuran (Ascending - Insertion)
Pilih : 3

=== Data Sneaker ===
1. Converse ChuckTaylor | Red 41 | 850000 | Bekas
2. Nike AirForce1 | White 42 | 1500000 | Baru
3. Adidas Yeezy350 | Black 43 | 4200000 | Baru
```

Gambar 4.6 Sorting Ukuran (Ascending / Insertion Sort)

```
=== MENU ===
1. Tampil Data
2. Sorting
3. Keluar
Pilih : 3
```

Gambar 4.7 Menu Keluar

5. Langkah-langkah GIT

5.1 GIT Init

```
PS C:\Users\MyASUS\OneDrive\文档\praktikum-apl\post-test\post-test-apl-5> git init
Initialized empty Git repository in C:/Users/MyASUS/OneDrive/文档/praktikum-apl/post-test/post-test-apl-5/.git/
```

5.2 GIT Add

```
PS C:\Users\MyASUS\OneDrive\文档\praktikum-apl\post-test\post-test-apl-5> git add .
```

5.3 GIT Commit

```
PS C:\Users\MyASUS\OneDrive\文档\praktikum-apl\post-test\post-test-apl-5> git commit -m "Done Posttest 5"
[master (root-commit) e033a4c] Done Posttest 5
2 files changed, 166 insertions(+)
create mode 100644 2509106125-CeciliaMarsya-POSTTEST-5.cpp
create mode 100644 2509106125-CeciliaMarsya-POSTTEST-5.exe
```

5.4 GIT Push

```
PS C:\Users\MyASUS\OneDrive\文档\praktikum-apl\post-test\post-test-apl-5> git push -u origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 8 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (4/4), 24.01 KiB | 1.14 MiB/s, done.
Total 4 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/cecilpua/praktikum-apl
  31f848a..6934d4a  main -> main
Writing objects: 100% (4/4), 24.01 KiB | 1.14 MiB/s, done.
Total 4 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/cecilpua/praktikum-apl
  31f848a..6934d4a  main -> main
  31f848a..6934d4a  main -> main
branch 'main' set up to track 'origin/main'.
```